

Web Accessibility

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Web Accessibility

- Web accessibility means that websites, tools, and technologies are designed and developed so that people with disabilities can use them.
- Web accessibility encompasses all disabilities that affect access to the Web, including: *auditory, cognitive, neurological, physical, speech, visual*
- Web accessibility also benefits people *without* disabilities, e.g., different form factors, different input modes, older people, situational limitation (e.g., noisy environments), slow Internet connection.
- Web accessibility is **required by law** in many situations.

Why is It Matters?

- About 1 in 6 people worldwide lives with a significant disability (WHO).
- Everyone can face limitations at some point, whether permanent, temporary, situational, or due to aging.
- Accessibility features like captions, good contrast, and clear layouts make websites easier for all users.
- Many countries require accessibility by law, including the ADA in the USA, the European Accessibility Act in the EU, and Law No. 8 of 2016 in Indonesia.
- Accessible websites reach a wider audience, rank better in search engines, and reduce legal risk.
- The web was built to be universal, and everyone deserves equal access to information.

Making the Web Accessible

- The W3C Web Accessibility Initiative ([WAI](#)) develops technical specifications, guidelines, techniques, and supporting resources that describe accessibility solutions.
 - WAI-ARIA (Accessible Rich Internet Applications)
 - ATAG and UAAG (authoring tools and user agents)
- International standards for web accessibility:
 - Web Content Accessibility Guidelines (**WCAG 2.2**)
 - is also an ISO standard: **ISO/IEC 40500**

POUR Principle

The POUR Principles of Web Accessibility

The foundation of WCAG – and the blueprint for designing inclusive, user-friendly websites.



Perceivable

Users must be able to see or hear the content.



Operable

Navigation must work for everyone – mouse, keyboard, or voice.



Understandable

Content should be clear, predictable, and easy to follow.



Robust

The site should work across devices and assistive tech.

<https://thrivethemes.com/designing-for-web-accessibility/>

Best Practices

■ Content and Images

- Add descriptive alt text to every meaningful image, and use empty alt text (alt="") for purely decorative images.
- Provide captions for videos and transcripts for audio content.
- Write in clear, simple language and break long text into short paragraphs.
- Use descriptive link text such as "Download the annual report" instead of "Click here."
- Make content easy to see and hear



Best Practices

■ Structure and Semantics

- Use proper HTML elements like `<header>`, `<nav>`, `<main>`, `<button>`, and `<footer>` instead of generic `<div>` tags.
- Organize content with headings in a logical order (H1, then H2, then H3) without skipping levels.
- Set the page language with the `lang` attribute, such as `<html lang="en">`.
- Give every page a unique and descriptive title.

Best Practices

■ Visual Design

- Maintain a color contrast ratio of at least 4.5:1 for normal text and 3:1 for large text.
- Never rely on color alone to convey meaning, so add icons, text, or patterns as well.
- Make sure Text is readable and understandable
- Make sure text can be resized up to 200% without breaking the layout.
- Design responsive layouts that work on all screen sizes and orientations.

Best Practices

- **Keyboard and Navigation**

- Make sure every interactive element can be reached and used with any input modalities (including keyboard)
- Keep a visible focus indicator so users can see where they are on the page.
- Ensure the tab order follows a logical reading sequence.
- Add a "Skip to main content" link so keyboard users can bypass repeated menus.
- Ensure that user can easily navigate, find content, and determine where they are

Best Practices

■ Forms

- Connect every input field to a visible <label>.
- Show clear error messages that explain what went wrong and how to fix it.
- Mark required fields clearly, using text and not only an asterisk or color.
- Group related fields with <fieldset> and <legend>.

Best Practices

■ **Interactive Elements and Media**

- Use ARIA attributes only when native HTML cannot do the job, because the first rule of ARIA is to prefer native elements.
- Avoid auto-playing audio or video, or give users an easy way to pause it.
- Avoid content that flashes more than three times per second, as it can trigger seizures.
- Give users enough time to read and complete tasks, and allow them to extend time limits.

Best Practices

- Content is compatible with current and future user assistive tools.
- Some of assistive technology
 - Screen Readers
 - Magnifiers
 - Speech Recognition
 - Switch Devices
 - Braille Display